

Практикум E3

188 marks, 55 blocks, nine techniques. Every question in this topic is the same trade in reverse of E2: you are handed a function and you give back another function. Which rule you reach for is settled by the shape of what you were handed, and that decision takes two seconds and costs the whole question if you get it wrong.

О практикуме

Тема calculus.differentiation — 98 блоков и 350 баллов, вдвое больше верхней границы. Она разрезана поимённо на три части, и списки лежат в blocks карточки, как это сделано у A5 и A6. Доли 0.537 / 0.374 / 0.089 не подобраны на глаз: это 188, 131 и 31 балл из 350.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-e3-differentiation

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Practicum E3: differentiating

188 marks, 55 blocks, nine techniques. Every question in this topic is the same trade in reverse of E2: you are handed a function and you give back another function. Which rule you reach for is settled by the *shape* of what you were handed, and that decision takes two seconds and costs the whole question if you get it wrong.

Material. The part of `calculus.differentiation` in the AA HL archive where the answer is the derivative itself, sessions May 2021 — November 2025. The whole topic is 350 marks — twice what a practicum may hold — so it is cut in three by what the question asks for: *find $f'(x)$* here, *find the tangent* in E4, *find the acceleration* in E9.

This practicum is in English, like B2 to B5, E1 and E2. The checks speak

whichever language the notebook asks them to, and this one asks for English in the setup cell.

The one thing to carry out of here.

Reach for the rule by looking at the **written shape**, not at what the function is called. A bracket with something inside it that is not x — chain. Two things multiplied, both moving — product. Something on top and something on the bottom, both moving — quotient. Everything else is the power rule and the table.

And the thing that follows from it: **most of the marks in this topic are not the differentiation.** Sections 1–5 of the archive carry 71 marks between them; sections 6–9 carry 117. What the archive pays for is the algebra that gets you from your line to the printed one, and the reading of a derivative you already have.

Where the calculator sits. 73% of the marks formally carry one, and about 15 of the 188 actually need one. This is the

sixth topic in a row where that number is an illusion, and the widest gap of the six. Paper 3 in particular looks like calculator territory and is nothing of the sort: $\frac{45\sqrt{5}}{4}$, $-\frac{r^2}{y^3}$, $\frac{2\sqrt{3}}{9}$, $2|a|$.

How the checks work here. Three of them are new.

`verify_derivative(label, got, f)` differentiates the function from the question and compares — as functions, not as strings, so any equivalent arrangement passes. When your answer is wrong it does more than say so. It rebuilds the standard slips out of that same function — the chain factor dropped, the product done as $u'v'$, the quotient numerator the wrong way round, the power not lowered — and if what you wrote matches one of them, it tells you which one. Neither the right answer nor the wrong ones are stored anywhere: both are worked out from f .

`verify_stationary(label, got, f, domain)` takes a list of points, checks that the

derivative really is zero at each one **and** that the second coordinate is the value of f there, then scans the interval for points you missed.

`verify_constants(label, got, unknowns, conditions)` is for the questions where letters sit inside the function. There is nowhere to substitute the answer, so it substitutes into the *conditions the question states* — the asymptote, the point on the curve, the zero gradient — and names the one that fails.

```

import sys
sys.path.append('.')          # from
practicum/calculus to practicum/kit.py
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
Rational, sqrt, pi, E, diff, Abs

language('en')                # this
notebook is in English, and so are the checks

a, b, c, d, h, m, n, p, q, r = symbols('a b c
d h m n p q r')
alpha, theta = symbols('alpha theta')

print('ready; sympy', sp.__version__)
print('a derivative:      ', 2/(2 - x)**3)
print('with a letter in it: ', n*x**(n - 1))
print('an exact value:    ', 45*sqrt(5)/4)
print('a pair of points:   ', [(0, 1),
(pi/2, -3)])

```

Map of techniques

#	Technique	What the function looks like	First move
1	Power rule, term by term	a sum of powers of x	drop each power by one
2	The table	$\sec, \csc, \cot, \ln, e^x$	look it up, keep the minus
3	Chain rule	something inside a bracket	outside first, then times the inside
4	Product rule	two moving factors multiplied	write $u'v + wv'$ before computing
5	Quotient rule	moving top over moving bottom	$\frac{u'v - wv'}{v^2}$, in that order
6	To the printed line	<i>show that</i> $f'(x) = \dots$	common denominator, then factor
7	Second and higher	$f'', d^2y/dx^2$	tidy the first one <i>for</i> the second
8	A letter in it	a, k, m, n, r in the answer	treat it as a number, keep the modulus
9	Read the derivative	<i>where is the gradient zero</i>	solve $f' = 0$, then substitute back

The ladder goes by what the question asks for.

Rungs 1-5 — you are choosing the rule. Seventy-one of the 188 marks, and the choice is made by looking, not by computing. It is the part everyone drills, and it is worth the least.

Rungs 6-8 — you are finishing. The algebra to the printed line, the second derivative, the letter that will not go away. Fifty-one marks, and the differentiation in them is usually the easy half.

Rungs 9 — you are reading. Fifty-one marks on its own, all of it after the derivative already exists. Solve $f' = 0$; put the roots back into f ; look at signs. The lost marks here are not calculus at all — a coordinate given as one number, an interval read short, *decreasing* solved as an equation.

The table you must know cold. It is in the formula booklet, and looking it up

costs the seconds in which you were meant to recognise the shape.

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \csc x = -\csc x \cot x$$

$$\frac{d}{dx} \cot x = -\csc^2 x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

Part I — choosing the rule

Theory 1. The shape decides, and you decide before you write

Four rules cover everything in this topic, and which one applies is a property of how the function is **written**, not of what it is.

$$\underbrace{x^4 - 3x^3 + 3x}$$

a sum: term by term

$$\underbrace{e^{x^2+1}}$$

something inside: chain

$$\underbrace{e^{2x}(3x-4)}$$

two moving factors: product

$$\frac{3x+2}{\underbrace{4x^2-1}}$$

moving over moving: quotient

Three things are worth saying before any of the rules.

A constant is a constant even when it is a letter. In $x^3 - 3cx + 2$ the c is a parameter: $-3cx$ differentiates to $-3c$,

and the 2 disappears. People who differentiate c lose the question, not the mark.

A bottom without x in it is not a quotient. $\frac{2500-1000 \tan \theta}{3}$ is $\frac{1}{3}$ times

something — the quotient rule on it is legal and three times the work.

Expand first when expanding is cheap. $x(2-x)$ is a product, but writing $2x - x^2$ takes one second and removes the rule entirely. $x^n(a-x)^n$ is also a product and expanding it is out of the question — the difference is whether the expansion is finite and short.

Task 1 ● — term by term, twice

May 2021 TZ1 Paper 1 Q4(a), 1 mark

May 2025 TZ1 Paper 3 Q1(c)(i), 2 marks

(a) Consider $f(x) = -(x-h)^2 + 2k$, where $h, k \in \mathbb{R}$. Find $f'(x)$.

(b) Find $G'(t)$ for

$$G(t) = \frac{1}{16}t + \frac{3}{16}t^2 + \frac{5}{16}t^3 + \frac{7}{16}t^4.$$

```

qla = ...          # f'(x)
qlb = ...          # G'(t)

verify_derivative('la', qla, -(x - h)**2 +
2*k)
verify_derivative('lb', qlb, t/16 + 3*t**2/16
+ 5*t**3/16 + 7*t**4/16, var=t)

```

Theory 2. The table, and the two minus signs in it

$$\frac{d}{d\theta} \sec \theta = \sec \theta \tan \theta$$

$$\frac{d}{d\theta} \csc \theta = -\csc \theta \cot \theta$$

$$\frac{d}{d\theta} \cot \theta = -\csc^2 \theta$$

sec is the odd one out: it is the only one of the three without a minus. csc and cot both carry one, and in this archive the missing minus on csc is the single most common lost mark of the whole section — because it is the minus that lets $\frac{dL}{d\alpha} = 0$ have a solution at all. If your equation has no solution, look there first.

A second habit worth having: **check the sign against the picture.** A cost that must have a minimum has a derivative

that changes sign; a population model that is decreasing has $f' < 0$ everywhere. Thirty seconds of that catches more errors than re-differentiating does.

Task 2 — sec and tan in one line

November 2025 TZ3 Paper 1 Q8(b), 3 marks

Astrid walks across a beach at 0.8 ms^{-1} and then jogs along the promenade at 1.2 ms^{-1} ; the time she takes, in seconds, is

$$T = 500 \sec \theta + \frac{2500 - 1000 \tan \theta}{3}, \quad 0 < \tan \theta \leq \frac{5}{2}.$$

Find $\frac{dT}{d\theta}$.

q2 = ...

```
verify_derivative('2', q2, 500*sec(theta) +
(2500 - 1000*tan(theta))/3,
var=theta)
```

Theory 3. The chain rule, and the factor that goes missing

$$\frac{d}{dx}F(u(x)) = F'(u(x)) \cdot u'(x)$$

The reliable tell is not the shape of the formula but what evaluating it would make you do: if computing $f(2)$ forces you to work out something *inside* first, that inside comes back out as a multiplier.

$$e^{x^2+1} \rightarrow 2x$$

$$e^{x^2+1}$$

$$\sqrt{r^2 - x^2} \rightarrow \frac{-x}{\sqrt{r^2 - x^2}}$$

$$\ln(\cot x) \rightarrow \frac{-\csc^2 x}{\cot x}$$

The whole inside gets differentiated, constants included. For $\sqrt{r^2 - k^2 x^2}$ the inside derivative is $-2k^2 x$, and the k^2 stays. It is not being differentiated; it is coming along.

Rewriting first is usually cheaper than the quotient rule. $\frac{1}{(2-x)^2}$ is $(2-x)^{-2}$, and one chain rule beats one quotient

rule every time. Watch the two minus signs though: $-2(2-x)^{-3} \cdot (-1)$ comes out **positive**.

A chain of three is still one rule, applied three times. $\sin(\sin(\sin x))$ differentiates to $\cos(\sin(\sin x)) \cos(\sin x) \cos x$: one cosine per layer, and the layers peel from the outside in.

Task 3 — inside, then out

November 2022 Paper 1 Q1, 4 marks

May 2024 TZ2 Paper 3 Q1(a)(i), 2 marks

(a) The function g is defined by $g(x) = e^{x^2+1}$, where $x \in \mathbb{R}$. Find $g'(-1)$.

(b) Find an expression for $f'(x)$, where $f(x) = \frac{1}{(2-x)^2}$, $x \neq 2$.

```
G = exp(x**2 + 1)
q3a = ...          # g'(-1), exact
q3b = ...          # f'(x)

verify_exact('3a', q3a, diff(G, x).subs(x,
-1))
verify_derivative('3b', q3b, 1/(2 - x)**2)
```


Theory 4. Product and quotient, and the two ways to lose them

$$(uv)' = u'v + uv' \qquad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

The product has two terms, and the method mark is for having two.

Writing $u'v'$ is not a slip in the arithmetic, it is a different (wrong) rule, and mark schemes give it nothing.

The quotient numerator does not commute. $\frac{uv' - u'v}{v^2}$ is exactly minus the right answer, which is worse than a random error: it looks plausible, it has the right size, and it takes the rest of the question with it.

Factorise before you go on. Almost every product-rule answer in this archive is wanted in a factorised form, because the *next* part of the question sets it to zero:

$$\begin{aligned} nx^{n-1}(a-x)^n + x^n \cdot n(a-x)^{n-1}(-1) &= nx^{n-1}(a-x)^{n-1}[(a-x) - x] \\ &= nx^{n-1}(a-2x)(a-x)^{n-1}. \end{aligned}$$

The bracket is $(a-x) - x$ and not $(a-x) + x$ because of the -1 from the chain rule

inside the product rule. That -1 is three of the five marks.

Task 4 — a product with a letter in the exponent

May 2021 TZ2 Paper 3 Q1(c), 5 marks

Consider $f_n(x) = x^n(a - x)^n$, where $a \in \mathbb{R}^+$ and $n \in \mathbb{Z}^+$, $n > 1$.

Show that $f'_n(x) = nx^{n-1}(a - 2x)(a - x)^{n-1}$.

```
q4 = ...
```

```
verify_derivative('4', q4, x**n*(a - x)**n,  
params={n: (2, 3, 5)})
```

Task 5 — two quotients, one with a letter

May 2021 TZ1 Paper 2 Q11(b), 3 marks

May 2025 TZ3 Paper 2 Q5(a), 3 marks

(a) Find an expression for $f'(x)$, where $f(x) = \frac{3x + 2}{4x^2 - 1}$.

(b) Find an expression for $f'(x)$ in terms of a , where $f(x) = \frac{(2x + a)^3}{(x + 5)^2}$, $x \neq -5$, $a \in \mathbb{R}^+$

.

```
q5a = ...          # for  $(3x + 2)/(4x^2 - 1)$   
q5b = ...          # for  $(2x + a)^3/(x + 5)^2$ ,  
in terms of a
```

```
verify_derivative('5a', q5a, (3*x +  
2)/(4*x**2 - 1))  
verify_derivative('5b', q5b, (2*x + a)**3/(x  
+ 5)**2, params={a: (1, 3, 7)})
```

Part II — finishing

Theory 5. The printed line is the question

Half the marks in this topic sit after the differentiation is done. A *show that* question hands you the destination, and what it is testing is the route: a common denominator, a factor taken out, a sign moved.

Two rules from the mark schemes, both worth money.

Work in one direction. Starting from your expression and from the printed one and meeting in the middle is not accepted. Start at yours and arrive.

Arrive all the way. $\frac{2x}{(2-x)^2} + \frac{2x^2}{(2-x)^3}$ is correct and is not the answer. The common denominator is $(2-x)^3$ — not $(2-x)^5$, because the first denominator divides the second — and then

$$\begin{aligned}\frac{2x(2-x) + 2x^2}{(2-x)^3} &= \frac{4x - 2x^2 + 2x^2}{(2-x)^3} \\ &= \frac{4x}{(2-x)^3}.\end{aligned}$$

The $-2x^2$ and the $+2x^2$ cancelling is the entire point of that question, and it is four marks.

Task 6 — the same destination, twice

*May 2024 TZ2 Paper 3 Q1(a)(ii) and (iii),
2 + 4 marks*

Let $f(x) = \frac{1}{(2-x)^2}$ and $g(x) = x^2$.

(a) Show that $f'(x)g'(x) = \frac{4x}{(2-x)^3}$.

(b) Show that $f(x)g'(x) + g(x)f'(x) = \frac{4x}{(2-x)^3}$.

*Same answer, twice the marks for (b).
The difference is one common denominator — and the coincidence that makes it work is what the rest of that Paper 3 is about.*

```

F, G = 1/(2 - x)**2, x**2
q6a = ...          # f'(x) g'(x)
q6b = ...          # f(x) g'(x) + g(x) f'(x)

verify_identity('6a', q6a, diff(F, x)*diff(G,
x))
verify_identity('6b', q6b, F*diff(G, x) +
G*diff(F, x))

```

Theory 6. The second derivative, and the form you leave the first in

Differentiate, then differentiate that. Two things make it harder than it sounds.

The form of f' decides how long f'' takes. For $f(x) = \sqrt{1+x}$, leaving $f'(x) = \frac{1}{2}(1+x)^{-1/2}$ gives f'' in one line. Leaving it as $\frac{1}{2\sqrt{1+x}}$ invites the quotient rule and three times the work, for the same answer.

An identity between derivatives can be differentiated as it stands. Once you have shown $g'' = 2(g' - g)$, you get $g''' = 2(g'' - g')$ and $g^{(4)} = 2(g''' - g'')$ for nothing — no further differentiation of g

at all. That is what *hence deduce* means, and it is two of the five marks.

Substitute back what you know about the curve. For $y = \sqrt{r^2 - x^2}$,

$$\frac{d^2y}{dx^2} = -\frac{y^2 + x^2}{y^3} = -\frac{r^2}{y^3},$$

because $x^2 + y^2 = r^2$. The tidy form is not cosmetic — the next part divides by it.

Task 7 — a relation between derivatives

May 2022 TZ1 Paper 1 Q12(c), 5 marks

The function g is defined by $g(x) = e^x \cos x$, where $x \in \mathbb{R}$.

(i) Show that $g''(x) = 2(g'(x) - g(x))$.

(ii) Hence, deduce that $g^{(4)}(x) = 2(g'''(x) - g''(x))$.

Enter $g''(x)$. Part (ii) needs no differentiation of g at all — see the solution.

```
q7 = ...
```

```
verify_derivative('7', q7, exp(x)*cos(x),  
order=2)
```

Task 8 — a pole round a corner, all the way to a number

*May 2023 TZ1 Paper 1 Q11(b)(i) and (c),
5 + 7 marks*

A pole of length L is carried round a corner between a passageway of width $\frac{3}{4}$ m and a room of width 6 m, at an angle α to the wall, $0 < \alpha < \frac{\pi}{2}$:

$$L = \frac{3}{4} \sec \alpha + 6 \csc \alpha.$$

(a) Find $\frac{dL}{d\alpha}$.

(b) Find $\frac{d^2L}{d\alpha^2}$.

(c) When $\alpha = \arctan 2$, show that $\frac{d^2L}{d\alpha^2} = \frac{45\sqrt{5}}{4}$.


```

L = 3*sec(alpha)/4 + 6*cosec(alpha)
q8a = ...           # dL/d(alpha)
q8b = ...           # d^2L/d(alpha)^2
q8c = ...           # its exact value at alpha =
arctan 2

verify_derivative('8a', q8a, L, var=alpha)
verify_derivative('8b', q8b, L, var=alpha,
order=2)
verify_exact('8c', q8c, simplify(diff(L,
alpha, 2).subs(alpha, arctan(2))))

```

Theory 7. A letter in the function

The rules do not change. What changes is the ending: there is no number to arrive at, so you have to decide when the expression is simplified enough — and two habits decide it for you.

Check on a value. If the answer is supposed to be in terms of m , put $m = 2$ into both the original and your answer and differentiate numerically. Ten seconds, and it catches a dropped factor every time.

Keep the modulus. $k_{\max} = 2|a|$, not $2a$. A parabola opening downwards curves

exactly as hard as the one opening upwards, and the question that comes next — *which of p and q is more curved* — is answerable only if the modulus survived. Writing $2a$ gets one wrong answer and then a second one.

Task 9 — the curvature of a parabola

May 2025 TZ1 Paper 3 Q2(b)(i) and (ii), 1 + 2 marks

The curvature k of a twice-differentiable function f is

$$k(x) = \frac{|f''(x)|}{\left(1 + (f'(x))^2\right)^{\frac{3}{2}}}.$$

For the family $h(x) = ax^2 + bx + c$, $a \neq 0$, it is given that

$$\begin{aligned} k(x) &= \frac{2|a|}{\left(1 + (2ax + b)^2\right)^{\frac{3}{2}}}, & k'(x) \\ &= -\frac{12a|a|(2ax + b)}{\left(1 + (2ax + b)^2\right)^{\frac{5}{2}}}. \end{aligned}$$

(a) By solving $k'(x) = 0$, find the value of x where $k_{\max}$ occurs.

(b) Determine an expression for $k_{\max}$, in terms of a only.

```
q9a = ...          # the x where k_max happens
q9b = ...          # k_max, in terms of a

verify_constants('9a', [q9a], [x],
                 [('the numerator of k'(x)
vanishes there', 2*a*x + b)])
verify_exact('9b', q9b, 2*Abs(a))
```

Part III — reading the derivative

Theory 8. Fifty-one marks after the calculus is over

Rung 9 is the largest single block in the topic, and it needs no technique that is not already above. The derivative exists — you found it in the part before, or the question printed it — and what is being asked is what it says.

$f'(x) = 0 \rightarrow$ where the tangent is flat

$f'(x) < 0 \rightarrow$ decreasing

$f''(x) > 0 \rightarrow$ concave-up

Four ways the marks actually go, all of them non-mathematical.

"Coordinates" means two numbers.

Solving $f'(x) = 0$ gives you the x 's; each one has to go back into f . In the November 2022 question that is four of the seven marks.

The interval in the question is the interval. $e^{\cos 2x}$ on $-\frac{\pi}{4} \leq x \leq \frac{5\pi}{4}$ has three

flat points, not two. Read the upper limit, then check whether the next solution fits — do not assume either way.

"Decreasing" is an inequality, not an equation. Finding the critical values is the method mark; writing the intervals between them is the answer mark. And when there are two intervals, giving one of them scores the same as giving none.

A given derivative is the starting point. When the question says *the derivative of f is given by $f'(x) = \dots$* , concave-up needs f'' , which means differentiating what you were handed — not f , which you do not have.

Task 10 — find it, then read it

November 2022 Paper 1 Q10(b), 7 marks

The function f is defined by $f(x) = \cos^2 x - 3 \sin^2 x$, $0 \leq x \leq \pi$.

(i) Find $f'(x)$.

(ii) Hence find the coordinates of the points on the graph of $y = f(x)$ where

$$f'(x) = 0.$$

```
q10a = ...          # f'(x)
q10b = [...]        # the points, each written
as (x, y)

verify_derivative('l0a', q10a, cos(x)**2 -
3*sin(x)**2)
verify_stationary('l0b', q10b, cos(x)**2 -
3*sin(x)**2, domain=(0, pi))
```

Task 11 — two inequalities from one given derivative

May 2025 TZ2 Paper 2 Q3, 3 + 3 marks

The derivative of a function f is given by $f'(x) = 4 + 2x - 3e^x$, where $x \in \mathbb{R}$.

(a) Find the values of x for which f is decreasing.

(b) Find the values of x for which the graph of f is concave-up.

This is the one question in the practicum where the calculator earns its place: the two critical values in (a) are $-1.73554\dots$ and $0.517999\dots$, and nothing is going to give them to you exactly.

```

q11a = ...      # where f is decreasing
q11b = ...      # where the graph is
concave-up

verify_param_set('11a', q11a,
                  lambda val: bool((4 + 2*x -
3*exp(x)).subs(x, val).evalf() < 0),
                  var=x, window=(-8, 4),
tol=Rational(1, 100))
verify_param_set('11b', q11b,
                  lambda val: bool((2 -
3*exp(x)).subs(x, val).evalf() > 0),
                  var=x, window=(-8, 4),
tol=Rational(1, 100))

```

Theory 9. When the letters are inside the function

Sometimes the unknowns are not the answer to a derivative but constants buried in the function, and the question pins them with facts about the curve. There is nothing to differentiate your way to; there is a system to build.

Each sentence of the question is one equation, and the trick is knowing which kind:

- **a vertical asymptote at $x = 1 \rightarrow$ the denominator is zero there;**

- **the curve passes through** $(2, 1) \rightarrow$ substitute the point;
- **a local minimum at** $(2, 1) \rightarrow$ the derivative is zero there, which for a quotient means the **numerator** of $\frac{dy}{dx}$ is zero there.

The mark scheme for the question below adds a warning worth carrying around: *an incorrect numerator may lead to a correct equation* — and in that case the mark is not given. Arriving at the right answer by the wrong route is specifically not credited here.

Task 12 **— three sentences, three equations**

May 2024 TZ2 Paper 2 Q9, 8 marks

Consider the curve

$$y = \frac{x - 4}{ax^2 + bx + c},$$

where a , b and c are non-zero constants. The curve has a local minimum point at

$(2, 1)$ and a vertical asymptote with equation $x = 1$.

Find the values of a , b and c .

```
a_, b_, c_ = symbols('a_ b_ c_')
Y = (x - 4)/(a_*x**2 + b_*x + c_)
q12 = [...]          # a, b, c in that order

verify_constants('12', q12, [a_, b_, c_], [
    ('the denominator is zero at the
    asymptote x = 1',
     (a_*x**2 + b_*x + c_).subs(x, 1)),
    ('the curve passes through (2, 1)',
     Eq(Y.subs(x, 2), 1)),
    ('and the gradient there is zero, because
    it is a minimum',
     diff(Y, x).subs(x, 2)),
    ])
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which rule you would reach for first**. That is the decision the whole topic turns on, and on the paper you get about five seconds of it.

code	technique
power	power rule, term by term
table	a standard derivative: sec, cosec, cot, ln
chain	something inside a bracket
product	two moving factors multiplied
quotient	moving over moving
t	
form	the answer is printed; the work is algebra
higher	a second or higher derivative
letter	the answer has to be in terms of a letter
read	the derivative exists; say what it means

1. Find an expression for $f'(x)$ where

$$f(x) = \frac{3x + 2}{4x^2 - 1}.$$

2. Find $\frac{dy}{dx}$ for $y = \sqrt{r^2 - x^2}$.

3. Find $f'(x)$ for $f(x) = e^{2x}(3x - 4)$.

4. Find the values of x for which the graph of f is concave-up, given $f'(x) = 4 + 2x - 3e^x$.

5. Find $\frac{d^2y}{dx^2}$ for $y = x^4 - 3x^3 + 3x$.

6. Find $\frac{dT}{d\theta}$ for $T = 500 \sec \theta + \frac{2500 - 1000 \tan \theta}{3}$.

7. Write down an expression for $f'(x)$ where $f(x) = x^3 - 3cx + 2$.
8. Show that $f(x)g'(x) + g(x)f'(x) = \frac{4x}{(2-x)^3}$.
9. Determine an expression for $k_{\max}$ in terms of a only.
10. Find the coordinates of the points on $y = e^{\cos 2x}$ where the gradient is zero.
1. Show that $S'_3(x) = \cos(\sin(\sin x)) \cos(\sin x) \cos x$.
2. Prove that $f'(\sqrt{A})$ is independent of A , where $f(x) = \frac{x(x^2 - A)}{x^2 + A}$.

```
answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}
```

```
trigger_check(answers, {1: 'd861ee362e38', 2:
'9414886b1ebf', 3: 'a8792157cb4f', 4:
'3316348dbadf', 5: 'e5ff8d2b6809', 6:
'0d4fc4a78d37', 7: '07d715edb696', 8:
'07397d633f25', 9: 'f80105efab6d', 10:
'3316348dbadf', 11: '9414886b1ebf', 12:
'd861ee362e38'})
```

On the timer

May 2023 TZ2 Paper 1 Q10(a), (b) and (c) — 8 marks. Ten minutes.

Top of the notebook covered, paper and pen, formula booklet allowed.

A circle with equation $x^2 + y^2 = 9$ has centre $(0, 0)$ and radius 3. A triangle PQR is inscribed in the circle with vertices at $P(-3, 0)$, $Q(x, y)$ and $R(x, -y)$, where Q and R are variable points in the first and fourth quadrants respectively.

(a) For point Q, show that $y = \sqrt{9 - x^2}$.

(b) Hence, find an expression for A , the area of triangle PQR, in terms of x .

(c) Show that $\frac{dA}{dx} = \frac{9 - 3x - 2x^2}{\sqrt{9 - x^2}}$.

Enter the answers to (b) and (c). Part (a) is one line on paper.

date	time	notes

```
qt_a = ...      # A, the area, in terms of x
qt_b = ...      # dA/dx

verify_identity('timer (b)', qt_a, (x +
3)*sqrt(9 - x**2))
verify_derivative('timer (c)', qt_b, (x +
3)*sqrt(9 - x**2))
```



Solutions

Task 1 — term by term, twice

(a) The $2k$ is a constant and goes; the chain rule on $(x - h)^2$ gives $2(x - h)$:

$$f'(x) = -2(x - h) = 2h - 2x.$$

A1. One mark, both forms accepted.

(b)

$$\begin{aligned} G'(t) &= \frac{1}{16} \\ &+ \frac{6}{16}t \\ &+ \frac{15}{16}t^2 \\ &+ \frac{28}{16}t^3 \\ &= \frac{1}{16} \\ &+ \frac{3}{8}t \\ &+ \frac{15}{16}t^2 \\ &+ \frac{7}{4}t^3. \end{aligned}$$

A1A1. Unsimplified sixteenths are accepted.

The reason that question exists: G is a probability generating function, and $G'(1) = \frac{1+6+15+28}{16} = \frac{50}{16}$, which is $E(M)$. Differentiating a list of probabilities produces its mean.

Task 2 — sec and tan in one line

$$\frac{dT}{d\theta} = 500 \sec \theta \tan \theta - \frac{1000}{3} \sec^2 \theta.$$

A1A1A1 — one for $\sec \theta \tan \theta$, one for $\sec^2 \theta$, one for keeping the $\frac{1}{3}$.

Two things to notice. The constant $\frac{2500}{3}$ vanishes entirely. And the $\frac{1}{3}$ has to survive onto the tangent term: the denominator belongs to the whole second fraction, not to the 2500 alone. That is the third mark and the one most often dropped.

Task 3 — inside, then out

(a) Chain rule with $u = x^2 + 1$:

$$g'(x) = 2x e^{x^2+1}, \quad g'(-1) = -2e^2.$$

A1A1A1A1. $(-1)^2 + 1 = 2$, not 0 — the square eats the sign, and the exponent is 2.

(b) Rewrite as $(2 - x)^{-2}$:

$$f'(x) = -2(2 - x)^{-3} \cdot (-1) = \frac{2}{(2 - x)^3}.$$

M1A1. Two minus signs, both real, and they cancel. Forgetting the inner -1 gives $-\frac{2}{(2-x)^3}$ — the right size and the wrong sign, which is the worst kind of error because nothing downstream looks odd.

The quotient rule also works here and takes twice as long. Rewriting a reciprocal as a negative power is the single cheapest habit in this topic.

Task 4 — a product with a letter in the exponent

$u = x^n$, $v = (a - x)^n$, and v needs the chain rule:

$$\begin{aligned} f'_n(x) &= nx^{n-1}(a - x)^n + x^n \cdot n(a - x)^{n-1} \cdot (-1). \end{aligned}$$

Take out $nx^{n-1}(a - x)^{n-1}$ — the lowest power of each:

$$\begin{aligned} &= nx^{n-1}(a - x)^{n-1}[(a - x) - x] \\ &= nx^{n-1}(a - 2x)(a - x)^{n-1}. \quad \mathbf{AG} \end{aligned}$$

M1A1A1M1A1. Five marks and three of them are the factorisation. The bracket is $(a - x) - x$ because of the chain-rule -1 ; with $+x$ it would collapse to a and the whole of parts (d) to (h) would come out wrong.

Notice what the factorised form buys: $f'_n(x) = 0$ now reads off as $x = 0, \frac{a}{2}, a$ with no work at all. That is the next two marks.

Task 5 — two quotients, one with a letter

(a) $u = 3x + 2$, $v = 4x^2 - 1$:

$$\begin{aligned}f'(x) &= \frac{3(4x^2 - 1) - (3x + 2)(8x)}{(4x^2 - 1)^2} \\&= \frac{12x^2 - 3 - 24x^2 - 16x}{(4x^2 - 1)^2} \\&= \frac{-12x^2 - 16x - 3}{(4x^2 - 1)^2}.\end{aligned}$$

M1A1A1. The numerator does not factorise over the integers; this is the final form.

(b) $u = (2x + a)^3$, $v = (x + 5)^2$, and both need the chain rule:

$$\begin{aligned}f'(x) &= \frac{6(2x + a)^2(x + 5)^2 - 2(2x + a)^3(x + 5)}{(x + 5)^4}.\end{aligned}$$

Cancel one $(x + 5)$ and take out $2(2x + a)^2$:

$$\begin{aligned}&= \frac{2(2x + a)^2[3(x + 5) - (2x + a)]}{(x + 5)^3} \\&= \frac{2(2x + a)^2(x - a + 15)}{(x + 5)^3}.\end{aligned}$$

M1A1A1. The mark scheme accepts the unsimplified version — and the next part

of that question needs $f'(1) = \tan 70^\circ$, which on the tidy form is one substitution and on the untidy one is a page. The two values that come out are $a = 2.73$ and $a = 15.0$.

Task 6 — the same destination, twice

(a) $f'(x) = \frac{2}{(2-x)^3}$ and $g'(x) = 2x$, so the product is $\frac{4x}{(2-x)^3}$. **A1A1** — two marks for a multiplication, because the paper is establishing this expression once and using it twice.

(b)

$$\begin{aligned}
 & fg' \\
 & \quad + gf' \\
 & \quad + \frac{2x^2}{(2-x)^3} \\
 & = \frac{2x(2-x) + 2x^2}{(2-x)^3} \\
 & = \frac{4x - 2x^2 + 2x^2}{(2-x)^3} \\
 & = \frac{4x}{(2-x)^3}. \quad \mathbf{AG}
 \end{aligned}$$

M1A1A1A1.

The common denominator is $(2 - x)^3$ because $(2 - x)^2$ divides it. Using $(2 - x)^5$ is not wrong, it is just three lines longer and one of those lines usually goes astray.

What this pair of functions is for: $fg' + gf' = f'g'$, that is, the product rule and the wrong "rule" happen to agree. Part (b) of that Paper 3 then rearranges the coincidence into $\frac{f'}{f} = \frac{g'}{g' - g}$, which is a differential equation in disguise.

Task 7 — a relation between derivatives

$$\begin{aligned} g'(x) &= e^x \cos x - e^x \sin x, & g''(x) \\ &= -2e^x \sin x. \end{aligned}$$

And $2(g' - g) = 2(-e^x \sin x) = -2e^x \sin x = g''$
. AG M1A1A1

For (ii), do **not** differentiate g two more times. Differentiate the identity:

$$g''' = 2(g'' - g'),$$

$$g^{(4)} = 2(g''' - g'').$$

A1A1

An equality between functions stays an equality when both sides are differentiated, and *hence deduce* is telling you so. Two marks, no work.

That identity is also how E2 builds the Maclaurin series for $e^x \cos x$: put $x = 0$ into it and the derivatives at zero become a numerical recursion.

Task 8 — a pole round a corner

(a)

$$\frac{dL}{d\alpha} = \frac{3}{4} \sec \alpha \tan \alpha - 6 \csc \alpha \cot \alpha.$$

A1A1. The minus on the $\csc$ term is what makes $\frac{dL}{d\alpha} = 0$ solvable; without it the two terms have the same sign and there is no stationary point at all.

Setting it to zero, in sines and cosines:

$$\begin{aligned}
\frac{3 \sin \alpha}{4 \cos^2 \alpha} &= \frac{6 \cos \alpha}{\sin^2 \alpha} \\
&\implies \sin^3 \alpha \\
&= 8 \cos^3 \alpha \\
&\implies \tan^3 \alpha \\
&= 8 \\
&\implies \tan \alpha \\
&= 2.
\end{aligned}$$

The cube root of a real number is unique, so $\tan \alpha = 2$ and no $\pm$.

(b) Product rule on each term:

$$\begin{aligned}
&\frac{d^2 L}{d\alpha^2} \\
&= \frac{3}{4} (\sec \alpha \tan^2 \alpha + \sec^3 \alpha) + 6 (\csc \alpha \cot^2 \alpha + \csc^3 \alpha).
\end{aligned}$$

A1A1A1A1.

(c) $\tan \alpha = 2$ means a right triangle with legs 2 and 1 and hypotenuse $\sqrt{5}$, so $\sec \alpha = \sqrt{5}$, $\csc \alpha = \frac{\sqrt{5}}{2}$, $\cot \alpha = \frac{1}{2}$:

$$\begin{aligned}
& \frac{3}{4} (4\sqrt{5} + 5\sqrt{5}) \\
& + 6 \left(\frac{\sqrt{5}}{8} + \frac{5\sqrt{5}}{8} \right) = \frac{27\sqrt{5}}{4} \\
& + \frac{18\sqrt{5}}{4} \\
& = \frac{45\sqrt{5}}{4}. \quad \mathbf{AG}
\end{aligned}$$

M1A1A1.

Draw the triangle. Going through $\alpha = \arctan 2$ on a calculator and back gives a decimal that will not turn into $\frac{45\sqrt{5}}{4}$, and this is a Paper 1.

Both terms come out positive, which is the point: $\frac{d^2L}{d\alpha^2} > 0$ makes $\alpha = \arctan 2$ a minimum of L . And the minimum of L is the longest pole that fits round the corner:

$$L_{\min} = \frac{3\sqrt{5}}{4} + 3\sqrt{5} = \frac{15\sqrt{5}}{4} \approx 8.39 \text{ m},$$

which is well short of 11.25 m — so the pole does **not** go round. The whole twenty-mark question exists for that one comparison, and it turns on L being

minimised: the tightest position is the one that decides.

Task 9 — the curvature of a parabola

(a) $k'(x) = 0$ needs the numerator to vanish, and $12a|a| \neq 0$, so $2ax + b = 0$:

$$x = -\frac{b}{2a}.$$

A1. The vertex — a parabola curves hardest exactly where it turns, which is worth a second's thought before you believe it.

(b) At that x the bracket $(2ax + b)$ is zero, so the denominator of k is 1:

$$k_{\max} = 2|a|.$$

M1A1.

The modulus is the mark. $p(x) = -2x^2 + 2x - 10$ and $q(x) = 2x^2 + 5x + 25$ both have $|a| = 2$, so both have $k_{\max} = 4$ and part (iv)'s answer is **C**. With $2a$ instead you would get -4 and 4 , conclude that q

curves more, and lose two questions from one dropped pair of bars.

Task 10 — find it, then read it

(i) $f(x) = \cos^2 x - 3 \sin^2 x$, so by the chain rule on each square

$$\begin{aligned} f'(x) &= -2 \cos x \sin x - 6 \sin x \cos x \\ &= -8 \sin x \cos x \\ &= -4 \sin 2x. \end{aligned}$$

A1A1. Rewriting with the double angle first — $f = \frac{1+\cos 2x}{2} - 3 \cdot \frac{1-\cos 2x}{2} = 2 \cos 2x - 1$ — gives the same answer in one line, and makes (ii) instant.

(ii) $\sin 2x = 0$ on $0 \leq x \leq \pi$ at $x = 0, \frac{\pi}{2}, \pi$, and back into f :

$$(0, 1), \quad \left(\frac{\pi}{2}, -3\right), \quad (\pi, 1).$$

M1A1A1A1A1. Five of the seven marks are here and four of them are the y -coordinates. The endpoints count — the interval is closed.

Task 11 — two inequalities from one given derivative

(a) Decreasing means $f'(x) < 0$, that is $4 + 2x - 3e^x < 0$. On the GDC the two critical values are $-1.73554\dots$ and $0.517999\dots$, and between them $f' > 0$ (at $x = 0$ it is 1). So

$$x \leq -1.74 \quad \text{or} \quad x \geq 0.518.$$

A1M1A1. The mark scheme accepts strict and non-strict alike, so the endpoints are not where the mark is. Giving only one of the two stretches is.

(b) Concave-up means $f''(x) > 0$, and $f''(x) = 2 - 3e^x$:

$$\begin{aligned} 2 - 3e^x > 0 &\implies e^x < \frac{2}{3} \\ &\implies x < \ln \frac{2}{3} (= -0.405). \end{aligned}$$

A1M1A1. One stretch this time, because f'' is decreasing everywhere.

The trap in (b) is differentiating f . You do not have f — the question gave you f' , and f'' is one differentiation of *that*.

Task 12 — three sentences, three equations

Vertical asymptote at $x = 1$. The denominator vanishes there:

$$a + b + c = 0.$$

The curve passes through $(2, 1)$.

$$\frac{2 - 4}{4a + 2b + c} = 1 \implies 4a + 2b + c = -2.$$

A local minimum at $x = 2$. The gradient is zero there, and for a quotient that means the numerator of $\frac{dy}{dx}$ is zero:

$$(ax^2 + bx + c) - (x - 4)(2ax + b) = 0 \text{ at } x = 2,$$

$$\begin{aligned}(4a + 2b + c) - (-2)(4a + b) &= 0 \\ \implies 12a + 4b + c &= 0.\end{aligned}$$

Solving the three:

$$a = 3, \quad b = -11, \quad c = 8.$$

M1A1A1M1(M1)A1M1A1. Eight marks, two of them calculus.

The mark scheme's warning is the thing to remember: *an incorrect numerator*

may lead to a correct equation — and then the mark is withheld. A right answer reached by a wrong route is specifically not credited on this question, which is unusual enough to be worth knowing exists.

On the timer

(a) Q is on the circle in the first quadrant, so $x^2 + y^2 = 9$ with $y > 0$ gives $y = \sqrt{9 - x^2}$. **A1**

(b) QR is vertical with length $2y = 2\sqrt{9 - x^2}$, and the horizontal distance from P to that line is $x - (-3) = x + 3$:

$$\begin{aligned} A &= \frac{1}{2} \cdot 2\sqrt{9 - x^2} \cdot (x + 3) \\ &= (x + 3)\sqrt{9 - x^2}. \end{aligned}$$

M1A1A1

(c) Product rule, with the chain rule inside:

$$\begin{aligned}
\frac{dA}{dx} &= \sqrt{9 - x^2} + (x + 3) \cdot \frac{-x}{\sqrt{9 - x^2}} \\
&= \frac{(9 - x^2) - x(x + 3)}{\sqrt{9 - x^2}} \\
&= \frac{9 - 3x - 2x^2}{\sqrt{9 - x^2}}. \quad \mathbf{AG}
\end{aligned}$$

M1A1M1A1

The $-2x^2$ needs both x^2 terms: one from $9 - x^2$ and one from $-x(x + 3)$. Losing either gives $9 - 3x - x^2$, which looks entirely reasonable and is not the printed line.

Part (d), for the record: $9 - 3x - 2x^2 = 0$ gives $x = \frac{3}{2}$, and the y -coordinate of R is $-\sqrt{9 - \frac{9}{4}} = -\frac{3\sqrt{3}}{2}$.

Key to the recognition drill

1 **quotient** — moving top over moving bottom. 2 **chain** — a bracket with x inside a square root. 3 **product** — two moving factors multiplied. 4 **read** — the derivative is given; concave-up is a statement about f'' . 5 **higher** — a second

derivative, asked for directly. 6 **table** — sec and tan, straight from the booklet. 7 **power** — a sum of powers, and c is a constant. 8 **form** — the answer is printed; the work is a common denominator. 9 **letter** — *in terms of a only*. 10 **read** — solve $f' = 0$, then substitute back for the coordinates. 11 **chain** — three nested sines, three cosine factors. 12 **quotient** — and then a substitution, but the rule is the quotient rule.

Three of these are worth arguing about, and the argument is the point.

2 could be power. $\sqrt{r^2 - x^2}$ is $(r^2 - x^2)^{1/2}$, so the power rule is involved — but the base is not x , and that is exactly what makes it **chain**. The power rule alone applies to x^n ; anything else inside needs its own derivative brought out.

5 could be power. It is: $\frac{d^2}{dx^2}(x^4 - 3x^3 + 3x)$ is two applications of the power rule and nothing else. It is **higher** because the thing being drilled is doing it twice without losing the constant on the way.

12 could be letter. The answer does have A in it — right up until it does not, which is the whole question. The rule you reach for is still the quotient rule; the letter is the punchline, not the technique.

Where the marks went, across the topic

technique	marks	share
Read the derivative	51	27%
Second and higher	29	15%
A letter in it	27	14%
Quotient rule	22	12%
Chain rule	18	10%
Product rule	14	7%
To the printed line	10	5%
The table	10	5%
Power rule, term by term	7	4%

Three things stand out.

Choosing the rule is worth a quarter of the topic. Rungs 1–5 carry 71 marks between them and rungs 6–9 carry 117.

The differentiation is the part everyone practises and the part that is worth least; the algebra afterwards and the reading of the result are worth more and get practised less. That is the same shape E2 had — building the series was worth less than spending it — and it is starting to look like a property of the whole calculus section rather than of either topic.

Reading the derivative is the largest block and needs no new technique.

Fifty-one marks, all of them after the calculus is over. The failures there are a coordinate given as one number, an interval read short, and *decreasing* solved as an equation — none of which is a failure of calculus.

Thirty-four of the fifty-five blocks are Paper 3. This topic does not appear as a five-mark question; it appears as a step inside a twenty-eight mark investigation, taken over and over. Two of them — May 2025 on curvature and November 2022

on surfaces of revolution — carry 39
marks between them.